

Module 4

Advanced Workflows

55 minutes

The hour

- 3 min — intro + worktrees
- 25 min — MCP mob (build `assign_route`)
- 10 min — worktrees + parallel agents
- 15 min — write a skill solo
- 2 min — wrap

15-minute break after.

MCP tool convention

Each tool: three exports plus one registration in `index.ts`.

Example (`list_vehicles`):

- `listVehiclesInputShape` — Zod input schema
- `listVehiclesDescription` — what the tool does and returns
- `listVehiclesHandler` — the work

Mirror this when adding new tools.

Description is pedagogy

The LLM reads it to decide *when* to call your tool.

Vague descriptions cause the agent to guess, misuse the tool, or hallucinate the response shape.

Highest-leverage line per tool. Invest here.

Building `assign_route` — decisions

1. **Schema** — what fields?
2. **Maintenance** — what if vehicle is in maintenance?
3. **Capacity** — what if `loadKg > capacityKg` ?
4. **Conflict** — what if vehicle is already on a route?
5. **Success** — if it succeeds, return `{ ok: true }` or updated state?
6. **Bidirectional** — what if `route.assignedVehicleId` isn't updated?

Worktrees + parallel agents

Multiple sessions — agents in different worktrees, independent file state.

Sub-agents — one session delegates scoped work via the Task tool.

Parallel is overhead. Default: one agent, one task, one session.

Write a skill

Pick one:

1. **Code review** in DispatchKit style
2. **PR description** in a fixed format
3. **Plan a feature** from a one-liner

Template: `modules/module-4/sample/skill-template/SKILL.md`

Lives at: `.claude/skills/<name>/SKILL.md`

Coming up: Module 5

When AI follows the plan but the code doesn't.

- Plan-first
- The golden rule
- Bidirectional sync
- Review a real PR

15-minute break.